UX/UI PROJECT · VACATION RENTAL PLATFORM

Cala

Usability evaluation, accessibility and inclusive design

Test plan, prototype evaluation and improvement proposals

Document	Academic project
Study object	Cala prototype: explore, details, reservation and confirmation
Date	October 3, 2026
Accessibility goal	WCAG 2.2, level AA

Summary

This project documents the Cala prototype inspection and designs an inclusive usability study. It examines journey comprehension, price transparency and barriers to access. The proposed protocol involves eight participants, privacy measures and two emerging technologies.

Scope of results: findings come from the supplied HTML, CSS and JavaScript. No participant sessions or browser execution checks have been conducted. No fabricated metrics or testimonials are presented.

1. Context, scope and objectives

Cala represents a vacation rental application through four screens: explore accommodations, consult the details, review the reservation and receive confirmation. The file contains low and high versions fidelity.

The only programmed behavior is the fidelity change using the setFi(mode) function. The Search, filters, cards, dates, and booking and payment actions are static representations. The four screens are displayed together; there is no functional navigation path.

Evaluation objectives

1. Check if you understand how to search for and select accommodation.
2. Verify that services, dates, guests and total price are identified.
3. Detect doubts about payment, conditions and confirmation.
4. Identify visual, auditory, motor and cognitive barriers.
5. Prioritize improvements and define how to verify their effectiveness.

Two stages of study

Stage	What allows you to evaluate	Limit
Current prototype	Comprehension, identification of information and intention of use	Pointing out a control is not equivalent to completing a reservation
Interactive version future	Task success, errors, time and use of assistive technologies	Requires implementing navigation and test controls

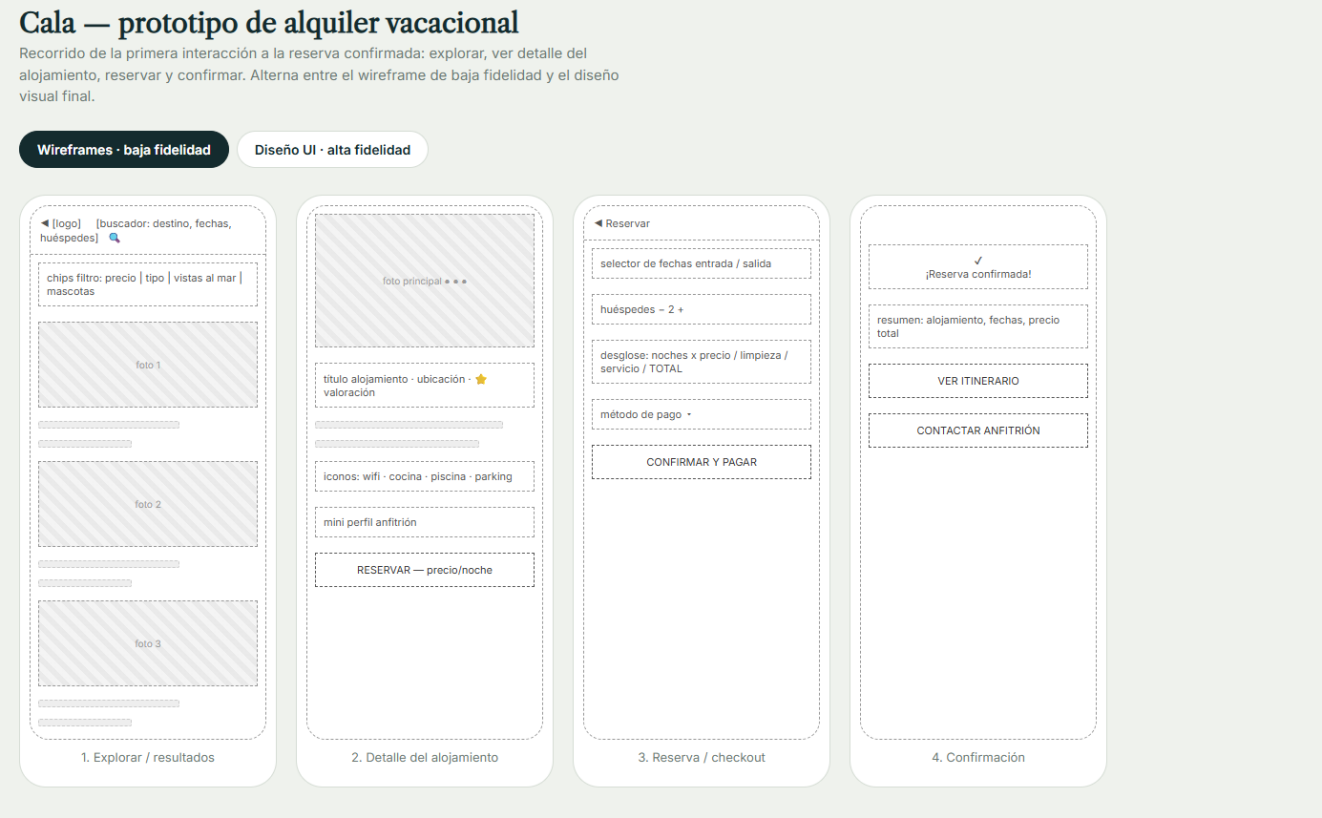
Project structure

2. Method and participants · 3. Tasks and indicators · 4. Development and analysis · 5. Findings and improvements · 6. Privacy · 7. Inclusive design · 8. Accessibility · 9. Emerging technologies · 10. Conclusions and annexes.

Visual presentation of the prototype

The following screenshots, contributed by the user, represent the two versions of the Cala interface. Both organize the tour in four screens: explore, accommodation details, reservation and confirmation.

Figure 1. Low fidelity wireframes



Representation of the structure, hierarchy and distribution of content using boxes, lines and markers of image. Source: screenshot provided by the user.

Visual presentation: interface design

Figure 2. High fidelity interface



Application of visual identity: colors, typography, cards, economic summary and actions planned in each screen. Source: screenshot provided by the user.

The screenshots document the appearance of the interface; do not prove by themselves that the actions of search, reservation or payment are functional nor that sessions have been held with participants.

2. Method, participants and sessions

An individual moderated test with observation and thinking aloud is proposed, followed by short questions. If verbalizing interferes with assistive technology or causes fatigue, retrospective questions. The moderator will facilitate the tasks without indicating where the solution is located.

The proposed sample includes eight adults with different ages, digital experience and support needs. It is a qualitative study: it is not intended to statistically represent the entire population. population.

Code	Planned profile
P01-P02	Two people with different digital experience, without declared support needs
P03	Person with low vision using magnification
P04	Blind person using screen reader
P05	Person with a motor disability who uses a keyboard or alternative input
P06	Deaf or hearing loss person
P07	Person with comprehension, attention or reading needs
P08	Older person with little digital experience

Profiles may overlap. You will ask what adaptations each person needs, without requiring diagnoses. You will be allowed to use your usual device and settings.

Phase	Indicative duration
Information, consent and settings	5 minutes
Context of use	5 minutes
Tasks	25 minutes
Assessment and final interview	10 minutes

A pilot session will be held first to adjust the script. Breaks and more time will be offered when be necessary. Duration will not be a condition of participation.

3. Tasks and success indicators

Understanding tasks for the current prototype

Task	Expected result
T1. Identify pool and pet admission	Recognize both services at Villa Cala Gat
T2. Explain how you would proceed to book	Identify the Reserve action
T3. Interpret the room	August 12-18, 2 guests and 6 nights
T4. Explain the full cost	€1,367: stay, cleaning and service
T5. Interpret the confirmation	Recognize accommodation, dates and total
T6. Locate subsequent actions	Identify itinerary and contact with the host

The moderator will present the high fidelity view. These tasks assess understanding and intention to use. The Static actions will be recorded as non-executable, not as completed functional tasks.

Indicators and internal goals

Indicator	Registration	Proposed objective
Understanding	Correct answer without help	At least 7 out of 8 per task
Full price	Identifies total and charges	8 out of 8 before confirming
Future Functional Success	Complete without help	At least 7 out of 8
Future reservation time	Simulated search to confirmation	Median less than 3 minutes
Perceived ease	Question from 1 to 7 after each task	Median of at least 5
Critical barriers	Blocks that prevent you from continuing	None pending

In the functional version you will try searching, filtering, opening an accommodation, modifying dates and guests, reviewing the cost and simulate the payment. Fictitious data will be used without charges. Goals are not normative standards; They will be reviewed after the pilot.

4. Development and analysis of tests

Preparation

The prototype, script, adaptations and consent will be checked. It will be explained: “We are evaluating the design; You can stop or ask for a break whenever you want.” Notifications will be disabled personal and fictitious scenarios will be used.

Log during tasks

The observer will record participant code, task, response, time when applicable, errors, help, barriers and relevant comments. You will classify the result as success without help, success with help, abandonment or not evaluable. The absence of functionality will be recorded as a limitation of the prototype.

Planned analysis

Incidents will be grouped by task and probable cause. Each problem will include evidence, participants affected, impact, proposal and subsequent verification. Individual counts will be reported and medium; Percentages will not be generalized with such a small sample.

Severity	Criterion
Criticism	Prevents you from completing an essential task or accessing your information
High	Causes major errors or requires help to continue
Medium	It produces doubts or effort, although it allows you to continue
Low	Affects comfort or presentation without impeding the task

The results with assistive technologies will be analyzed according to the barriers found. A longer time does not will automatically be attributed to the person's capacity. Design problems, limitations will be separated of the prototype and conditions of the session.

Execution status

Completed: reading and inspection of the code and review of economic consistency. Pending: sessions, task measurements, subjective assessment, browser testing and full audit. They don't exist yet satisfaction data or user testimonials.

5. Findings and proposals for improvement

Code finding	Impact and improvement	Priority
Search and reservation controls without logic	They prevent the flow from being evaluated. Implement navigation and dummy data.	Criticism
Tabs div with onclick, without keyboard or control semantics	They prevent standard keyboard access. Use native buttons with focus and accessible state.	High
Price per night until checkout	Difficult to compare the full cost. Show total for dates chosen from the list.	High
Dates, guests and payment static	They do not allow you to correct selections. Create labeled controls and validation.	High
Fixed heights and overflow:hidden	Risk of cuts when expanding. Allow growth or displacement and check visually.	High; check
No meta viewport	Risk of inappropriate mobile presentation. Add it and review adaptation.	Medium
No visible cancellation policy	Lack of information to decide. Show conditions before payment.	High
Photographs replaced by degraded	Limits confidence evaluation. Incorporate images representative and alternative.	Medium

The breakdown is arithmetically consistent: €210 × 6 nights + €45 cleaning + €62 service = €1,367. The confirmation keeps the same total.

Priorities are inspection estimates. Its frequency and real impact will be contrasted with participants. Correcting controls is the first step to be able to measure effectiveness and efficiency.

6. Data collection and privacy

Only participant code, approximate digital experience, requested adaptations, task results and comments. An observation sheet and a short questionnaire will be used. The screen or audio recording will be optional and will have separate authorization; rejecting it will not prevent participate.

Before the sessions, you will be informed about the person responsible, purpose, legal basis, recipients, conservation and exercise of rights. The GDPR requires minimization and transparency. The institution must specify these data before starting the study [3].

Measure	Application
Pseudonymization	Codes P01-P08; contacts and authorizations in a separate file
Fictitious data	Do not ask for personal accounts, card numbers or actual reservations
Limited access	Encrypted institutional folder, accessible only to the evaluation team
Minimum recording	Avoid notifications, personal desktop and information unrelated to the test
Proposed conservation	Recordings: 30 days; pseudonymized records: 90 days. Confirm before the study
Presentation of results	Aggregated data and de-identified quotes; review re-identification risks
External services	Do not send recordings to IA without evaluating treatment and informing previously

Pseudonymization is not equivalent to anonymization. Medical diagnoses will be avoided: it will be enough to know the necessary adaptations. If it is essential to process health data, the conditions will be reviewed applicable specifics.

The HTML loads fonts from Google Fonts. It is proposed to house them locally to reduce dependencies external. The inspection does not identify forms, analytics, or participant storage, but this does not replaces a review of network requests.

7. Inclusive design

Need	Adaptations and checks
visual	Screen reader, magnification, contrast, headers and image alternatives. Verify reading the total and confirmation.
Auditory	Written instructions and text communication. If videos are incorporated, provide subtitles and relevant alternatives.
motor	Full keyboard or alternative input, comfortable objectives and actions that do not require precision or drag.
Cognitive	Clear language, predictable steps, summary before payment, errors with a solution and the possibility of review without losing data.

Adaptations will be agreed with each participant. Instructions will be available in writing and They will allow breaks. Imposing a single form of interaction will be avoided.

Physical accommodation accessibility

The accessibility of the interface does not guarantee that of the accommodation. Cala should offer verified information on entrance without steps, elevator, widths and adapted bathroom. A generic “accessible” label does not Provides enough detail to decide.

Inclusive acceptance criteria

Essential tasks must be able to be completed with assistive technologies. The relevant content must remain available when expanding. The economic information will have a clear textual formulation and the Confirmations will not depend only on color or an icon.

After the improvements, the evaluation will be repeated with people who found barriers, when possible. If You will also verify that the changes do not generate difficulties in other forms of use.

8. Accessibility: WCAG 2.2 AA objective

Level AA is established as an objective because it includes relevant requirements for different forms of perception and interaction. Conformity requires satisfying all applicable criteria A and AA. The prototype cannot **be declared compliant currently: there are barriers in the code and complete checks are missing [1].**

Principle	Evaluation and improvement
Perceptible	Darken secondary text. Review image alternatives, enlargement and redistribution. Criteria relevant: 1.1.1, 1.3.1, 1.4.3, 1.4.4 and 1.4.10.
Operable	Replace tabs with accessible controls and verify keyboard, order and focus. Check size of objectives and non-hidden focus. Criteria: 2.1.1, 2.4.3, 2.4.7, 2.4.11 and 2.5.8.
Understandable	The Spanish language is declared. Explain conditions, errors and consequences; allow to correct before payment. Criteria: 3.1.1, 3.3.1, 3.3.2 and 3.3.4.
Robust	Use semantics, names, roles and accessible states. Announce confirmation whenever dynamic. Criteria: 4.1.2 and 4.1.3.

Contrast calculated from CSS colors

Text/background	Relationship	Normal text
#6E7B76 / #FFFFFF	4,41:1	insufficient
#6E7B76 / #EFF2ED	3,91:1	insufficient
#142B2E / #FFFFFF	14,84:1	Enough
#5B3E06 / #E3A62F	4,57:1	Enough

Criterion 1.4.3 requires 4.5:1 for normal text [2]. The calculations refer to these declared combinations; they do not test all rendered combinations or global conformance. review will be combined automatic, manual and supporting technologies.

9. Proposed emerging technologies

9.1. Multimodal conversational assistant with AI

It could make searching via text or voice easier and explain prices in plain language. I would help people with difficulties reading, data entry or understanding filters. It is presented as a future proposal, not as existing functionality.

Design conditions: Maintain conventional controls, display selections before applying them, and allow them to be corrected. Accommodation accessibility data must come from verified information. The assistant must not invent them or confirm payments autonomously.

Validation: compare comprehension and errors with and without assistance; check that people can Leave the conversation without losing your selections. Review the treatment of queries and avoid unnecessary conservation.

9.2. Speech recognition and automatic transcription

It could allow you to search for accommodations without writing and transcribing audio messages. Voice input can help people with motor difficulties; transcription can facilitate access to messages for people who are deaf or have hearing loss.

Design Conditions: Offer text as an alternative, ask for microphone permission, and allow editing and confirm what is recognized. Evaluate errors with different accents and forms of speech. Prefer processing local when feasible.

Validation: recording recognition errors, correction effort and transcription understanding. Economic operations should require explicit review.

Inclusive integration

These technologies should expand the available options without becoming a mandatory path. Its usefulness It will be validated with users and will not replace the basic accessibility of the site.

10. Conclusions and improvement plan

Cala presents a comprehensible conceptual sequence and a coherent economic breakdown. The inspection identifies static controls, keyboard barriers, insufficient contrast in secondary text, and risks of presentation, along with opportunities to show the total and conditions further in advance.

An inclusive protocol has been designed and the available inspection results have been documented. The actual sessions and the complete audit are pending. The report does not attribute participants results that have not yet been collected.

Order	Action	Check
1	Implement controls and simulated tour	Search and reservation tasks can be executed
2	Fix semantics, keyboard and contrast	Manual review and assistive technologies
3	Improve total price, conditions and edition	Understanding without help in the proposed tasks
4	Run pilot and eight sessions	Complete records, with documented limitations
5	Iterate and check again	No critical crashes; updated WCAG matrix

References

[1] W3C. Web Content Accessibility Guidelines (WCAG) 2.2.
https://www.w3.org/TR/WCAG22/

[2] W3C. Understanding Success Criterion 1.4.3: Contrast (Minimum).
Official Minimum Contrast Guide

[3] European Union. Regulation (EU) 2016/679. Articles 5, 6, 9, 13 and 32.
Official text of the GDPR in EUR-Lex

Evaluated material: Cala HTML file contributed by the user. References consulted on October 3 of 2026.

Annexes. Instruments for the sessions

A. Observation sheet

field	Content to be recorded
Participant/task	Pseudonymized code and task number
Settings	Device, browser and supports used
Result	Without help / with help / abandonment / not evaluable
Time and help	Duration when applicable and specific help
Evidence	Observed action or deidentified comment
Barrier and severity	Problem, consequence and priority
Ease	Rating from 1 to 7

B. Subsequent questions

1. What has been most difficult or confusing for you?
2. What amount do you think you would pay and what does it include?
3. What information would you need before confirming?
4. Have you encountered any barriers with your usual way of using the device?
5. From 1 to 7, how easy was this task?

C. Prior information and consent

Before participating, an accessible sheet will be delivered with the person responsible and contact, purpose, legal basis, recipients, deadlines and rights. It will be explained that there are no real payments, that participating is voluntary and that you can stop the session. Recording authorization will be collected separately. This annex is a script of preparation; must be specified before collecting data.

D. Accessibility matrix

For each applicable criterion, the following will be recorded: WCAG identifier, screen, method, evidence, result. (complies / does not comply / does not apply / pending), action and subsequent verification. AA will not be declared while there are non-compliances or pending applicable verifications.